



**THE
MATERIAL
THE MARKET
DOESN'T
HAVE YET.**

DARC — Discontinuous Advanced Resin Composite. A proprietary compression-moldable carbon fiber composite with a property stack that does not exist in any current production material. Developed through national laboratory collaboration. Eighteen months of founder-funded characterization.

CONFIRMED PROPERTIES

Lighter than aluminum and conventional carbon fiber prepreg. Exceptional impact toughness. Bulk electrical conductivity with a memristive conditioning effect — resistance floor not yet established. Through-thickness thermal conductivity. UV stable. Radio opaque. Complex monocoque geometries with molded-in hardware.

PROPERTIES UNDER INVESTIGATION

Structural health monitoring. Triboelectric energy harvesting. EMI shielding. Radar signature characteristics.

APPLICATIONS

Battery enclosures. Drone and autonomous system frames. Defense and aerospace structures. Any application where weight, toughness, thermal performance, and structural intelligence converge.

PROCESS AND SCALABILITY

Compression molding. Five minute press cycle. Well over one hundred parts per press per shift. Each press and mold set added multiplies output linearly. Material cost exceptionally low relative to target sale price.

VALIDATION AND NETWORK

National laboratory collaboration via NMSBA. Active characterization at NMSU Z-Lab and Physical Science Laboratory. New Mexico Tech engaged. Technical advisor Brad Jones, PhD. IP in patent preparation with national counsel. NMEDD grant recipient. Ecosystem: WESST, QStation, NM Startup Alliance, Arrowhead, APEX Accelerator, Southwest Composite Works.